Derivation of exact solution of 1D linear model problem describing waves along an equator with a uniformly deep ocean - JAN'10

The equations to solve: $\frac{\partial \zeta}{\partial t} + h \frac{\partial u}{\partial x} = 0$ and $\frac{\partial u}{\partial t} + U_0 \frac{\partial u}{\partial x} + g \frac{\partial \zeta}{\partial x} = -\lambda \cdot u + f$ with:

- uniform constant depth h (5000m),
- constant gravitational acceleration g (9.81m/s^2),
- some constant background current U_0 ([m/s]),
- constant linear friction coefficient $\lambda = g \frac{U_{typ}}{hC^2}$ ([1/s]),
- periodic tidal forcing/acceleration $f = F \cdot \exp(I \cdot (\omega \cdot t - k \cdot x))$ ([m/s^2]).

Periodic boundaries on a domain of length l ($2 \cdot \pi \cdot r_{earth} = 2\pi$*6800km = 42726km), hence in space solution and also forcing must be of the form $\exp(I \cdot k \cdot x)$,

with ($k = \frac{2 \cdot \pi \cdot n}{l} = \frac{n}{r_{earth}}$, n some integer > 0).

We could apply coordinate transformation $x = \frac{l \cdot \phi}{360}$, with ϕ angle in degrees.

=> $\frac{\partial \zeta}{\partial t} + h' \frac{\partial u}{\partial \phi} = 0$ and $\frac{\partial u}{\partial t} + U_0' \frac{\partial u}{\partial \phi} + g' \frac{\partial \zeta}{\partial \phi} = -\lambda \cdot u + f$ with: $h' = \frac{360\, h}{l}$, $U_0' = \frac{360\, U_0}{l}$, $g' = \frac{360\, g}{l}$, $f = F \cdot \exp(I \cdot (\omega \cdot t - k' \cdot \phi))$, $k' = \frac{2 \cdot \pi \cdot n}{360}$. Hm, perhaps not so useful.

```
> restart;
```

The equations

```
> cont := diff( zeta(x,t), t )                          + h * diff(  u(x,t),   x ) = 0;
  mom  := diff(  u(x,t),   t ) + U0 * diff( u(x,t), x ) + g * diff( zeta(x,t), x ) = - lambda * u(x,t) +
  F * exp( I*(omega*t - k*x) );
```

$$cont := \frac{\partial}{\partial t} \zeta(x,t) + h \left(\frac{\partial}{\partial x} u(x,t) \right) = 0$$

$$mom := \frac{\partial}{\partial t} u(x,t) + U0 \left(\frac{\partial}{\partial x} u(x,t) \right) + g \left(\frac{\partial}{\partial x} \zeta(x,t) \right) = -\lambda\, u(x,t) + F\, e^{I(\omega t - kx)}$$

The solution(s)

```
> zp := Z(t) * exp( I*(- k*x) ):
  up := U(t) * exp( I*(- k*x) ):
> contp := simplify( eval( subs( zeta(x,t)=zp, u(x,t)=up, cont ) ) / exp( I*(- k*x) ) );
  momp  := simplify( eval( subs( zeta(x,t)=zp, u(x,t)=up, mom  ) ) / exp( I*(- k*x) ) );
```

$$contp := \frac{d}{dt} Z(t) - I\,h\,U(t)\,k = 0$$

$$momp := \frac{d}{dt} U(t) - I\,U0\,U(t)\,k - I\,g\,Z(t)\,k = -\lambda\,U(t) + F\,e^{I\omega t}$$

```
> sol := dsolve( [contp,momp] );
```

$$sol := \left\{ U(t) = -\frac{1}{h\,k}\left(I\left(\frac{1}{2} I\left(I\lambda + U0\,k - \sqrt{-\lambda^2 + 2\,I\lambda\,U0\,k + U0^2\,k^2 + 4\,h\,k^2\,g}\right) e^{\frac{1}{2} I\left(I\lambda + U0k - \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C2 + \frac{1}{2} I\left(I\lambda + U0\,k + \sqrt{-\lambda^2 + 2\,I\lambda\,U0\,k + U0^2\,k^2 + 4\,h\,k^2\,g}\right) e^{\frac{1}{2} I\left(I\lambda + U0k + \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C1 - \frac{h\,k\,F\,\omega\,e^{I\omega t}}{h\,k^2\,g + I\lambda\,\omega + U0\,k\,\omega - \omega^2}\right)\right), Z(t) = e^{\frac{1}{2} I\left(I\lambda + U0k - \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C2 + e^{\frac{1}{2} I\left(I\lambda + U0k + \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C1 + \frac{I\,h\,k\,F\,e^{I\omega t}}{h\,k^2\,g + I\lambda\,\omega + U0\,k\,\omega - \omega^2} \right\}$$

```
> usol := subs( op(1,sol), up ) ;
  zsol := subs( op(2,sol), zp ) ;
```

$$usol := -\frac{1}{h\,k}\left(I\left(\frac{1}{2} I\left(I\lambda + U0\,k - \sqrt{-\lambda^2 + 2\,I\lambda\,U0\,k + U0^2\,k^2 + 4\,h\,k^2\,g}\right) e^{\frac{1}{2} I\left(I\lambda + U0k - \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C2 + \frac{1}{2} I\left(I\lambda + U0\,k + \sqrt{-\lambda^2 + 2\,I\lambda\,U0\,k + U0^2\,k^2 + 4\,h\,k^2\,g}\right) e^{\frac{1}{2} I\left(I\lambda + U0k + \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C1 - \frac{h\,k\,F\,\omega\,e^{I\omega t}}{h\,k^2\,g + I\lambda\,\omega + U0\,k\,\omega - \omega^2}\right) e^{-Ikx}\right)$$

$$zsol := \left(e^{\frac{1}{2} I\left(I\lambda + U0k - \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C2 + e^{\frac{1}{2} I\left(I\lambda + U0k + \sqrt{-\lambda^2 + 2I\lambda U0k + U0^2 k^2 + 4hk^2 g}\right) t}\,_C1 + \frac{I\,h\,k\,F\,e^{I\omega t}}{h\,k^2\,g + I\lambda\,\omega + U0\,k\,\omega - \omega^2} \right) e^{-Ikx}$$

```
> MTM[fortran](usol,uexa);
  MTM[fortran](zsol,zexa);
Warning, the following variable name replacements were made: ["cg", "cg1"] = [" C1", " C2"]
"     uexa = (0, -1) * ((0, 0.1D1 / 0.2D1) * ((0, 1) * lambda + U0 * k -
# csqrt(-lambda ** 2 + (0, 2) * lambda * U0 * k + U0 ** 2 * k ** 2
#+ 4 * h * k ** 2 * g)) * cexp((0, 0.1D1 / 0.2D1) * ((0, 1) * lambd
#a + U0 * k - csqrt(-lambda ** 2 + (0, 2) * lambda * U0 * k + U0 **
# 2 * k ** 2 + 4 * h * k ** 2 * g)) * t) * cg1 + (0, 0.1D1 / 0.2D1)
# * ((0, 1) * lambda + U0 * k + csqrt(-lambda ** 2 + (0, 2) * lambd
```

```
#a * U0 * k + U0 ** 2 * k ** 2 + 4 * h * k ** 2 * g)) * cexp((0, 0.
#1D1 / 0.2D1) * ((0, 1) * lambda + U0 * k + csqrt(-lambda ** 2 + (0
#, 2) * lambda * U0 * k + U0 ** 2 * k ** 2 + 4 * h * k ** 2 * g)) *
# t) * cg - 0.1D1 / (h * k ** 2 * g + (0, 1) * lambda * omega + U0
#* k * omega - omega ** 2) * h * k * F * omega * cexp((0, 1) * omeg
#a * t)) / h / k * cexp((0, -1) * k * x)
"
```

Warning, the following variable name replacements were made: ["cg", "cg1"] = [" C1", " C2"]

```
"    zexa = (cexp((0, 0.1D1 / 0.2D1) * ((0, 1) * lambda + U0 * k - csqr
#t(-lambda ** 2 + (0, 2) * lambda * U0 * k + U0 ** 2 * k ** 2 + 4 *
# h * k ** 2 * g)) * t) * cg1 + cexp((0, 0.1D1 / 0.2D1) * ((0, 1) *
# lambda + U0 * k + csqrt(-lambda ** 2 + (0, 2) * lambda * U0 * k +
# U0 ** 2 * k ** 2 + 4 * h * k ** 2 * g)) * t) * cg + (0, 1) / (h *
# k ** 2 * g + (0, 1) * lambda * omega + U0 * k * omega - omega **
#2) * h * k * F * cexp((0, 1) * omega * t)) * cexp((0, -1) * k * x)
"
```

```
> h := 5000:
  g := 9.81:
  l := 2*Pi*6800000:   # earth's circumference in meters
  period := 24*60*60: # number of seconds in a day
  k      := 2*Pi* nmode / l:
  omega := 2*Pi* nfreq / period:
  # nmode  := 1:
  # nfreq  := 1:
  # lambda := 0:
  # U0     := 0:
```

```
> # uexa := unapply( Re(usol), [nmode,nfreq,lambda,U0,_C1,_C2,F,x,t] );
  zexa := unapply( Re(zsol), [nmode,nfreq,lambda,U0,_C1,_C2,F,x,t] );
```

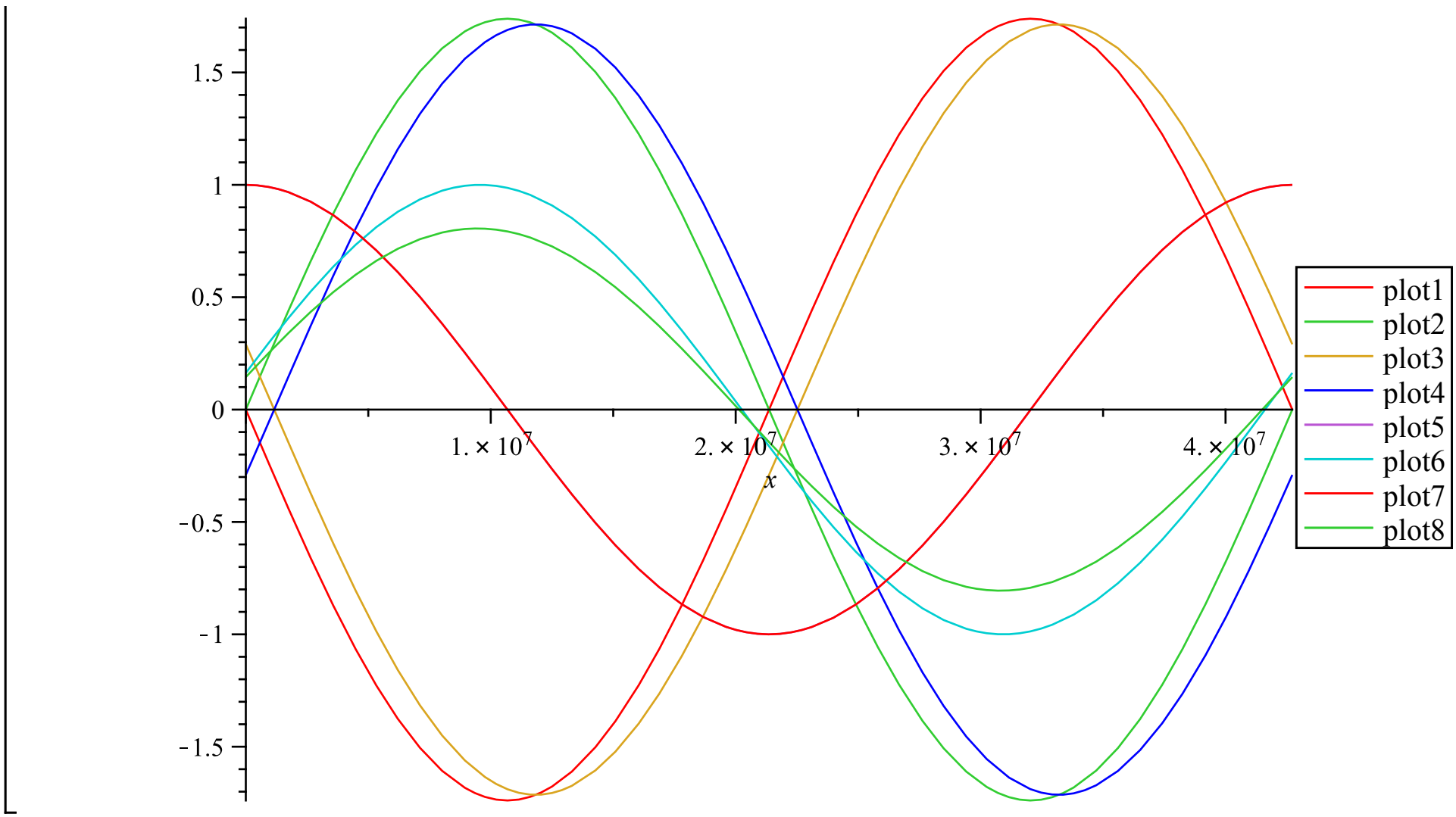
$$zexa := (nmode, nfreq, \lambda, U0, _C1, _C2, F, x, t)$$

$$\rightarrow \Re\left(\left(e^{\frac{1}{2} I \left(I\lambda + \frac{1}{6800000} U0\, nmode - \sqrt{-\lambda^2 + \frac{1}{3400000} I\lambda\, U0\, nmode + \frac{1}{46240000000000} U0^2\, nmode^2 + 4.243079584\, 10^{-9}\, nmode^2}\right) t} _C2\right.\right.$$

$$+ e^{\frac{1}{2} I \left(I\lambda + \frac{1}{6800000} U0\, nmode + \sqrt{-\lambda^2 + \frac{1}{3400000} I\lambda\, U0\, nmode + \frac{1}{46240000000000} U0^2\, nmode^2 + 4.243079584\, 10^{-9}\, nmode^2} \right) t} _C1$$

$$+ \frac{\frac{1}{1360} I\, nmode\, F\, e^{\frac{1}{43200} I \pi\, nfreq\, t}}{1.060769896\, 10^{-9}\, nmode^2 + \frac{1}{43200} I\lambda\pi\, nfreq + \frac{1}{293760000000} U0\, nmode\, \pi\, nfreq - \frac{1}{1866240000} \pi^2\, nfreq^2} \Bigg) e^{-\frac{1}{6800000} I\, nmode\, x} \Bigg)$$

```
> plot( [ zexa(1,1,   0   , 0 , 0,0,0.00001, x,   0    ),
          zexa(1,1,   0   , 0 , 0,0,0.00001, x,period/2),
          zexa(1,1,0.00001, 0 , 0,0,0.00001, x,   0    ),
          zexa(1,1,0.00001, 0 , 0,0,0.00001, x,period/2),
          zexa(1,1,   0   , 0 , 1,0,   0   , x,   0    ),
          zexa(1,1,   0   , 0 , 1,0,   0   , x,period/2),
          zexa(1,1,0.00001, 0 , 1,0,   0   , x,   0    ),
          zexa(1,1,0.00001, 0 , 1,0,   0   , x,period/2)
        ], x=0..l,
           legendstyle=[location=right], legend=[ "plot1", "plot2", "plot3", "plot4", "plot5", "plot6",
  "plot7", "plot8" ] );
```


1.5
1
0.5
0
-0.5
-1
-1.5
1. × 10^7
2. × 10^7
3. × 10^7
4. × 10^7
x
plot1
plot2
plot3
plot4
plot5
plot6
plot7
plot8